Pavan Ravindra

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EDUCATION

University of Maryland, College Park **Expected Graduation: May 2021**
B.S. in Biochemistry, B.S. in Computer Science *Cumulative GPA: 3.99/4.00*

- **Churchill Scholar (2021)** - given to 16 students in the US annually to fund a year of study at Cambridge University towards a master's degree in science, mathematics, or engineering
- **Goldwater Scholar (2020)** - the premier US scholarship for recognizing undergraduate research excellence in the natural sciences, engineering, and mathematics
- **Philip Merrill Presidential Scholar (2020)** - given to the top rising seniors in each UMD college based on academic performance
- **Honors College Citation (2019)** - Integrated Life Sciences Program
- **Banneker/Key Scholar (2017)** - the most prestigious 4-year merit scholarship offered to incoming UMD undergraduates - provides full college funding package
- Graduate coursework in Thermodynamics, Statistical Mechanics, and Nuclear Chemistry

RESEARCH EXPERIENCE

Tiwary Lab **2018-Present**
Undergraduate Researcher *University of Maryland, College Park*

- Advisor: Dr. Pratyush Tiwary
- Applies and develops algorithms to enhance sampling of molecular dynamics simulations
- Automatic Mutual Information Noise Omission (AMINO)
  - Developed, tested, and applied an algorithm that uses an information theoretic approach to identify a basis set of order parameters for use in reaction coordinate/collective variable construction for metadynamics simulations
  - **Publication:** Ravindra, P., Smith, Z., Tiwary, P. (2020). Automatic mutual information noise omission (AMINO): Generating order parameters for molecular systems. Molecular Systems Design & Engineering, 5(1), 339-348.
  - **Publication:** Smith, Z., Ravindra, P., Wang, Y., Cooley, R., Tiwary, P. (2020). Discovering Protein Conformational Flexibility Through Artificial Intelligence Aided Molecular Dynamics. The Journal of Physical Chemistry B.
  - **Presented Poster:** Ravindra P, Smith Z, and Tiwary P. 2019. Automatic mutual information noise omission (AMINO): generating order parameters for molecular systems. Poster session presented at: MolSSI Machine Learning and Chemistry Workshop; 2019 November 17.
- Using graphical models to analyze molecular simulations:
  - Applying recent advancements from computer science, mathematics, and statistics to explore applications of research in graphical models towards understanding molecular simulations

National Institutes of Health **Summer 2018**
Research Assistant *National Institute of Neurological Disorders and Stroke*

- Advisor: Dr. Jeffrey Smith
- Built computer simulations of the pre-Bötzinger complex, a group of pacemaker neurons that regulates intrinsic breathing patterns in mammals using a Hodgkin-Huxley formalism and a differential equation modelling software (Berkeley-Madonna)
- **Presented Poster:** Sarch G.H., Ravindra P., and Smith J. Computational Modeling of Respiratory Neural Circuits. Poster session presented at: NIH Summer Poster Day; 2018 August 9; Bethesda, MD.

TEACHING EXPERIENCE

Head Teaching Assistant - Computer Science **Fall 2019-Present**

CMSC330 - Organization of Programming Languages

- Leads a team of 35 TAs to organize one of the largest computer science courses at UMD (500+ students per semester)
- Organizes and coordinates weekly TA meetings to assign duties to other TAs
- Designs and writes projects, exams, and quizzes

Teaching Assistant - Computer Science **Spring 2019, Summer 2019**

CMSC330 - Organization of Programming Languages

- Planned and wrote weekly discussion activities for students
- Held office hours for students

Teaching Assistant - Chemistry **Spring 2019-Present**

CHEM242 - Organic Chemistry II Lab

- Directs weekly laboratory activities for a group of 18 undergraduate students
- Grades assignments and exams

EXTRACURRICULARS

Competitive Rubik's Cube Speedsolving

- Formerly ranked 4th in the world for fastest time solving a Rubik's cube
- Peak ranking of 2nd in the world for one-handed Rubik's cube solving
- Set 5 US national records

Rubik's Cube Competition Organizer

- President - University of Maryland Rubik's Cube Club
- Organizes competitions that attract 200+ international competitors
- Organized CubingUSA Northeast Championship 2019, the regional championship for the World Cube Association for the northeast US region

First Degree Black Belt and Instructor (Karate)

- Trained in martial arts for 10 years
- Volunteered to teach students of all skill/experience levels

Treasurer - University of Maryland Society of Microbiology

- Manages organization funds
- Assists in planning group meetings

Webmaster and Founding Member - University of Maryland NAMSA Organization

- INformative Association for Maryland Science Advancement
- Built and maintains the organization's website